(vi) Let A be a 3×3 matrix such that its eigen values are 2, -1 and λ. $(1,0,1)^T$ is the eigen vector corresponding to 2 and $(2,0,1)^T$ is the eigen vector corresponding to -1. If the second column of A is $(1,0,1)^T$, Find λ, A and $det(A)$. [5]

$A=[c_1\ c_2\ c_3]$ $\qquad c_2=[1\ 0\ 1]^t$

$A\cdot\begin{bmatrix}1\\0\\1\end{bmatrix}=2\begin{bmatrix}1\\0\\1\end{bmatrix} \Rightarrow c_1+c_2=[2\ 0\ 2]^t$

$A\begin{bmatrix}2\\0\\1\end{bmatrix}=-1\begin{bmatrix}2\\0\\1\end{bmatrix} \Rightarrow 2c_1+c_3=[-2\ 0\ -1]^t$

$c_1=[-4\ 0\ -3]^t$

$c_3=[6\ 0\ 5]^t$

$\Rightarrow A=\begin{bmatrix}-4 & 1 & 6\\ 0 & 0 & 0\\ -3 & 1 & 5\end{bmatrix}$

$|A|=0 \Rightarrow \lambda_1\cdot\lambda_2\cdot\lambda_3=0 \Rightarrow \lambda=0$

(12) Determine whether A is Diagonalizable?

$$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

If it is Diagonalizable, then Diagonalize A by finding a non-singular matrix S and a diagonal matrix D, such that $S^{-1}AS = D$ [5]

$$\begin{vmatrix} \lambda-2 & 1 & 1 \\ 1 & \lambda-2 & 1 \\ 1 & 1 & \lambda-2 \end{vmatrix} = 0 \Rightarrow (\lambda-2)[\lambda^2-4\lambda+4-1]$$

$$-1(\lambda-2-1)+1(1-\lambda+2)=0$$

$$\Rightarrow (\lambda-1)(\lambda-2)(\lambda-3)-(\lambda-3)-(\lambda-3)=0$$

$$\Rightarrow [\lambda-3][\lambda^2-3\lambda]=0 \Rightarrow (\lambda-3)^2.\lambda=0$$

$\sigma(A)=\{0,3,3\}$ $\quad AM(0)=1$, $AM(3)=2$

$N(A)$: $2v_1=v_2+v_3$, $2v_2=v_1+v_3$, $2v_3=v_1+v_2$ $\Rightarrow v_1=v_2=v_3 \Rightarrow [1,1,1]^t$

$\Rightarrow GM(0)=1$

$N(A-3I)$: $v_1+v_2+v_3=0 \Rightarrow v_3=-v_1-v_2$

$\Rightarrow [1,0,-1]^t, [0,1,-1]^t$

$GM(3)=2$

$\Rightarrow$ diagonalizable

$A=SDS^{-1}$ $\quad D=\begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ $\quad S=\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & -1 \end{bmatrix}$

(a) Let A be a real symmetric 3×3 matrix and $x = [x_1, x_2, x_3]^T$, such that $x^T x = (Ax)^T(Ax)$ for all $x \in \mathbf{R}^3$. Is it necessarily true that A is orthogonal? Justify. [3]

(b) Find the inverse of [2]

$$x^T x = x^T A^T A x$$
$$= x^T A^2 x$$

$$\Rightarrow x^T (I - A^2) x = 0 \quad \forall \; x \in \mathbb{R}^3$$

$$\Rightarrow I - A^2 = 0 \Rightarrow A^2 = I$$
$$\Rightarrow A = A^{-1}$$
$$\Rightarrow A^T = A^{-1}$$ Yes ✓

20. Let $A \in \mathbb{C}^{n\times n}$ be a Hermitian matrix. Prove that any eigenvalue of A is real. Further, let λ_1 and λ_2 be distinct eigenvalues of A with corresponding eigenvectors v_1 and v_2. Prove that v_1, v_2 are orthogonal. [4]

Sol Let λ be an eigenvalue of A with an eigenvector v. Then $Av = \lambda v$. Then

$A = A^*$

$A\vartheta = \lambda\vartheta$

$\vartheta^* A^* = \vartheta^* \bar{\lambda}$

$\Rightarrow \vartheta^* A = \vartheta^* \bar{\lambda}$

$\Rightarrow \vartheta^* A \vartheta = \bar{\lambda}\, \vartheta^* \vartheta$

$\Rightarrow \lambda \vartheta^* \vartheta = \bar{\lambda}\, \vartheta^* \vartheta$

$\Rightarrow \lambda = \bar{\lambda} \Rightarrow \lambda$ is real

$A\vartheta_1 = \lambda_1 \vartheta_1$

$A\vartheta_2 = \lambda_2 \vartheta_2$

$\langle A\vartheta_1, \vartheta_2 \rangle = A\vartheta_1 \vartheta_2^* = \lambda_1 \langle \vartheta_1, \vartheta_2 \rangle$

$\langle \vartheta_1, A\vartheta_2 \rangle = \vartheta_1 (A\vartheta_2)^* = \bar{\lambda}_2 \langle \vartheta_1, \vartheta_2 \rangle = \lambda_2 \langle \vartheta_1, \vartheta_2 \rangle$

also since $A^* = A$; $\langle A\vartheta_1, \vartheta_2 \rangle = \langle A^* \vartheta_1, \vartheta_2 \rangle = \langle \vartheta_1, A\vartheta_2 \rangle$

$\Rightarrow \lambda_1 \langle \vartheta_1, \vartheta_2 \rangle = \lambda_2 \langle \vartheta_1, \vartheta_2 \rangle \Rightarrow \langle \vartheta_1, \vartheta_2 \rangle = 0$

4. Let $a_1, \ldots, a_n \in \mathbb{C}$. Let A be the $n \times n$ matrix whose first row has all entries as 1 and whose kth row has entries $a_1^{k-1}, \ldots, a_n^{k-1}$ in that order. Show that $\det(A) = (-1)^{n(n-1)/2} \Pi_{1 \le i < j \le n}(a_i - a_j)$.

$$A = \begin{bmatrix} a_1^0 & a_2^0 & a_3^0 \ldots & a_n^0 \\ a_1^1 & a_2^1 & a_3^1 \ldots & a_n^1 \\ \vdots & \vdots & \vdots & \vdots \\ a_1^n & a_2^n & a_3^n \ldots & a_n^n \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 0 & 0 & \ldots & 0 \\ a_1 & a_2 - a_1 & a_3 - a_1 & & a_n - a_1 \\ \vdots & & & & \\ a_1^n & a_2^n - a_1^n & a_3^n - a_1^n & & a_n^n - a_1^n \end{vmatrix}$$

$$= \begin{vmatrix} a_2 - a_1 & a_3 - a_1 & \ldots & a_n - a_1 \\ a_2^2 - a_1^2 & a_3^2 - a_1^2 & \ldots & a_n^2 - a_1^2 \\ \vdots & & & \\ a_2^n - a_1^n & a_3^n - a_1^n & \ldots & a_n^n - a_1^n \end{vmatrix}$$

$$= (a_2 - a_1)(a_3 - a_1) \ldots (a_n - a_1) \begin{vmatrix} 1 & 1 & \\ a_2 + a_1 & a_3 + a_1 & \\ a_2^2 + a_1 a_2 + a_1^2 & a_3^2 + a_3 a_1 + a_1^2 & \\ \vdots & \vdots & \\ & & \end{vmatrix}$$

$$A = \begin{bmatrix} 2 & -1 \\ -2 & 1 \\ 4 & -2 \end{bmatrix}_{3\times 2}$$

$$AA^* = \begin{bmatrix} 5 & -5 & 10 \\ -5 & 5 & -10 \\ 10 & -10 & 20 \end{bmatrix} \qquad A^*A = \begin{bmatrix} 24 & -12 \\ -12 & 6 \end{bmatrix} \qquad \sigma(A^*A) = \{0, 30\}$$

$$\Rightarrow \Sigma = \begin{bmatrix} \sqrt{30} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A^*A v_1 = 30 v_1$$

$$\Rightarrow v_1 : B(R(A^*A - 30I)) \Rightarrow \text{Basis of } \begin{bmatrix} -6 & -12 \\ -12 & -24 \end{bmatrix}$$

$$\Rightarrow v_1 = \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

$$v_2 : N(A^*A) \Rightarrow v_2 = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} \qquad \Rightarrow V = \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$$

$$u_1 : Av_1/s_1 = \begin{bmatrix} -1/\sqrt{6} \\ 1/\sqrt{6} \\ -2/\sqrt{6} \end{bmatrix}$$

$$u_2 : N(AA^*) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

$$U = \begin{bmatrix} -1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \\ -2/\sqrt{6} & 0 & -1/\sqrt{3} \end{bmatrix}$$

$$Bx_i = \begin{bmatrix} A^* u_i \\ A v_i \end{bmatrix} = \begin{bmatrix} \lambda v_i \\ \lambda u_i \end{bmatrix}$$

$$A^* u_i = \lambda v_i$$
$$A v_i = \lambda u_i$$

$$u_i = A v_i / s_i \qquad v_i = A^* u_i / s_i$$

eigenvalues of B = singular values of A